通用方法

Element Software

NetApp
May 31, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/element-software/api/reference_element_api_getapi.html on May 31, 2022. Always check docs.netapp.com for the latest.

目录

通用方法 1
了解更多信息 1
GetAPI 1
GetAsyncResult 10
GetCompleteStats 13
GetLimits 13
GetOrigin 16
GetRawStats 17
ListAsyncResults 17

通用方法

常用方法是用于检索有关存储集群， API 本身或正在进行的 API 操作的信息的方法。

- GetAPI
- GetAsyncResult
- GetCompleteStats
- GetLimits
- GetOrigin
- GetRawStats
- ListAsyncResults

了解更多信息

- "NetApp SolidFire 资源页面"
- "早期版本的 NetApp SolidFire 和 Element 产品的文档"

GetAPI

您可以使用 `GetAPI` 方法获取可在系统中使用的所有 API 方法和受支持的 API 端点的列表。

Parameters

此方法没有输入参数。

返回值

此方法具有以下返回值：

Name	Description	Type
version	此软件版本支持的所有 API 方法的列表，其中 <version> 是此系统正在运行的当前软件版本。	string array
当前版本	存储集群软件的当前版本。	string
支持的版本	系统支持的所有 API 端点的列表。	string array

请求示例

此方法的请求类似于以下示例：

```
{
  "method": "GetAPI",
  "params": {},
  "id" : 1
}
```

响应示例

此方法返回类似于以下示例的响应：

```
{
"id": 1,
    "result": {
        "12.0": [
            "AbortSnapMirrorRelationship",
            "AddAccount",
            "AddClusterAdmin",
            "AddDrives",
            "AddIdpClusterAdmin",
            "AddInitiatorsToVolumeAccessGroup",
            "AddKeyServerToProviderKmip",
            "AddLdapClusterAdmin",
            "AddNodes",
            "AddVirtualNetwork",
            "AddVolumesToVolumeAccessGroup",
            "BreakSnapMirrorRelationship",
            "BreakSnapMirrorVolume",
            "CancelClone",
            "CancelGroupClone",
            "CheckPingOnVlan",
            "CheckProposedCluster",
            "CheckProposedNodeAdditions",
            "ClearClusterFaults",
            "CloneMultipleVolumes",
            "CloneVolume",
            "CompleteClusterPairing",
            "CompleteVolumePairing",
            "CopyVolume",
            "CreateBackupTarget",
            "CreateClusterInterfacePreference",
            "CreateClusterSupportBundle",
            "CreateGroupSnapshot",
            "CreateIdpConfiguration",
            "CreateInitiators",
```

```
            "CreateKeyProviderKmip",
            "CreateKeyServerKmip",
            "CreatePublicPrivateKeyPair",
            "CreateQoSPolicy",
            "CreateSchedule",
            "CreateSnapMirrorEndpoint",
            "CreateSnapMirrorEndpointUnmanaged",
            "CreateSnapMirrorRelationship",
            "CreateSnapMirrorVolume",
            "CreateSnapshot",
            "CreateStorageContainer",
            "CreateSupportBundle",
            "CreateVolume",
            "CreateVolumeAccessGroup",
            "DeleteAllSupportBundles",
            "DeleteAuthSession",
            "DeleteAuthSessionsByClusterAdmin",
            "DeleteAuthSessionsByUsername",
            "DeleteClusterInterfacePreference",
            "DeleteGroupSnapshot",
            "DeleteIdpConfiguration",
            "DeleteInitiators",
            "DeleteKeyProviderKmip",
            "DeleteKeyServerKmip",
            "DeleteQoSPolicy",
            "DeleteSnapMirrorEndpoints",
            "DeleteSnapMirrorRelationships",
            "DeleteSnapshot",
            "DeleteStorageContainers",
            "DeleteVolume",
            "DeleteVolumeAccessGroup",
            "DeleteVolumes",
            "DisableAutoip",
            "DisableBmcColdReset",
            "DisableClusterSsh",
            "DisableEncryptionAtRest",
            "DisableIdpAuthentication",
            "DisableLdapAuthentication",
            "DisableSnmp",
            "EnableAutoip",
            "EnableBmcColdReset",
            "EnableClusterSsh",
            "EnableEncryptionAtRest",
            "EnableFeature",
            "EnableIdpAuthentication",
            "EnableLdapAuthentication",
```

```
            "EnableSnmp",
            "GetAccountByID",
            "GetAccountByName",
            "GetAccountEfficiency",
            "GetActiveTlsCiphers",
            "GetAsyncResult",
            "GetBackupTarget",
            "GetBinAssignmentProperties",
            "GetClientCertificateSignRequest",
            "GetClusterCapacity",
            "GetClusterConfig",
            "GetClusterFullThreshold",
            "GetClusterHardwareInfo",
            "GetClusterInfo",
            "GetClusterInterfacePreference",
            "GetClusterMasterNodeID",
            "GetClusterSshInfo",
            "GetClusterState",
            "GetClusterStats",
            "GetClusterStructure",
            "GetClusterVersionInfo",
            "GetCompleteStats",
            "GetConfig",
            "GetCurrentClusterAdmin",
            "GetDefaultQoS",
            "GetDriveHardwareInfo",
            "GetDriveStats",
            "GetFeatureStatus",
            "GetFipsReport",
            "GetHardwareConfig",
            "GetHardwareInfo",
            "GetIdpAuthenticationState",
            "GetIpmiConfig",
            "GetIpmiInfo",
            "GetKeyProviderKmip",
            "GetKeyServerKmip",
            "GetLdapConfiguration",
            "GetLimits",
            "GetLldpInfo",
            "GetLoginBanner",
            "GetLoginSessionInfo",
            "GetNetworkConfig",
            "GetNetworkInterface",
            "GetNodeFipsDrivesReport",
            "GetNodeHardwareInfo",
            "GetNodeStats",
```

```
            "GetNtpInfo",
            "GetNvramInfo",
            "GetOntapVersionInfo",
            "GetOrigin",
            "GetPendingOperation",
            "GetProtectionDomainLayout",
            "GetQoSPolicy",
            "GetRawStats",
            "GetRemoteLoggingHosts",
            "GetSSLCertificate",
            "GetSchedule",
            "GetSnapMirrorClusterIdentity",
            "GetSnmpACL",
            "GetSnmpInfo",
            "GetSnmpState",
            "GetSnmpTrapInfo",
            "GetStorageContainerEfficiency",
            "GetSupportedTlsCiphers",
            "GetSystemStatus",
            "GetVirtualVolumeCount",
            "GetVolumeAccessGroupEfficiency",
            "GetVolumeAccessGroupLunAssignments",
            "GetVolumeCount",
            "GetVolumeEfficiency",
            "GetVolumeStats",
            "InitializeSnapMirrorRelationship",
            "ListAccounts",
            "ListActiveAuthSessions",
            "ListActiveNodes",
            "ListActivePairedVolumes",
            "ListActiveVolumes",
            "ListAllNodes",
            "ListAsyncResults",
            "ListAuthSessionsByClusterAdmin",
            "ListAuthSessionsByUsername",
            "ListBackupTargets",
            "ListBulkVolumeJobs",
            "ListClusterAdmins",
            "ListClusterFaults",
            "ListClusterInterfacePreferences",
            "ListClusterPairs",
            "ListDeletedVolumes",
            "ListDriveHardware",
            "ListDriveStats",
            "ListDrives",
            "ListEvents",
```

```
            "ListFibreChannelPortInfo",
            "ListFibreChannelSessions",
            "ListGroupSnapshots",
            "ListISCSISessions",
            "ListIdpConfigurations",
            "ListInitiators",
            "ListKeyProvidersKmip",
            "ListKeyServersKmip",
            "ListNetworkInterfaces",
            "ListNodeFibreChannelPortInfo",
            "ListNodeStats",
            "ListPendingActiveNodes",
            "ListPendingNodes",
            "ListProtectionDomainLevels",
            "ListProtocolEndpoints",
            "ListQoSPolicies",
            "ListSchedules",
            "ListServices",
            "ListSnapMirrorAggregates",
            "ListSnapMirrorEndpoints",
            "ListSnapMirrorLuns",
            "ListSnapMirrorNetworkInterfaces",
            "ListSnapMirrorNodes",
            "ListSnapMirrorPolicies",
            "ListSnapMirrorRelationships",
            "ListSnapMirrorSchedules",
            "ListSnapMirrorVolumes",
            "ListSnapMirrorVservers",
            "ListSnapshots",
            "ListStorageContainers",
            "ListSyncJobs",
            "ListTests",
            "ListUtilities",
            "ListVirtualNetworks",
            "ListVirtualVolumeBindings",
            "ListVirtualVolumeHosts",
            "ListVirtualVolumeTasks",
            "ListVirtualVolumes",
            "ListVolumeAccessGroups",
            "ListVolumeStats",
            "ListVolumeStatsByAccount",
            "ListVolumeStatsByVirtualVolume",
            "ListVolumeStatsByVolume",
            "ListVolumeStatsByVolumeAccessGroup",
            "ListVolumes",
            "ListVolumesForAccount",
```

```
            "ModifyAccount",
            "ModifyBackupTarget",
            "ModifyClusterAdmin",
            "ModifyClusterFullThreshold",
            "ModifyClusterInterfacePreference",
            "ModifyGroupSnapshot",
            "ModifyInitiators",
            "ModifyKeyServerKmip",
            "ModifyQoSPolicy",
            "ModifySchedule",
            "ModifySnapMirrorEndpoint",
            "ModifySnapMirrorEndpointUnmanaged",
            "ModifySnapMirrorRelationship",
            "ModifySnapshot",
            "ModifyStorageContainer",
            "ModifyVirtualNetwork",
            "ModifyVolume",
            "ModifyVolumeAccessGroup",
            "ModifyVolumeAccessGroupLunAssignments",
            "ModifyVolumePair",
            "ModifyVolumes",
            "PurgeDeletedVolume",
            "PurgeDeletedVolumes",
            "QuiesceSnapMirrorRelationship",
            "RemoveAccount",
            "RemoveBackupTarget",
            "RemoveClusterAdmin",
            "RemoveClusterPair",
            "RemoveDrives",
            "RemoveInitiatorsFromVolumeAccessGroup",
            "RemoveKeyServerFromProviderKmip",
            "RemoveNodes",
            "RemoveSSLCertificate",
            "RemoveVirtualNetwork",
            "RemoveVolumePair",
            "RemoveVolumesFromVolumeAccessGroup",
            "ResetDrives",
            "ResetNetworkConfig",
            "ResetNode",
            "ResetSupplementalTlsCiphers",
            "RestartNetworking",
            "RestartServices",
            "RestoreDeletedVolume",
            "ResumeSnapMirrorRelationship",
            "ResyncSnapMirrorRelationship",
            "RollbackToGroupSnapshot",
```

```
            "RollbackToSnapshot",
            "SecureEraseDrives",
            "SetClusterConfig",
            "SetClusterStructure",
            "SetConfig",
            "SetDefaultQoS",
            "SetLoginBanner",
            "SetLoginSessionInfo",
            "SetNetworkConfig",
            "SetNtpInfo",
            "SetProtectionDomainLayout",
            "SetRemoteLoggingHosts",
            "SetSSLCertificate",
            "SetSnmpACL",
            "SetSnmpInfo",
            "SetSnmpTrapInfo",
            "SetSupplementalTlsCiphers",
            "Shutdown",
            "SnmpSendTestTraps",
            "StartBulkVolumeRead",
            "StartBulkVolumeWrite",
            "StartClusterPairing",
            "StartVolumePairing",
            "TestAddressAvailability",
            "TestConnectEnsemble",
            "TestConnectMvip",
            "TestConnectSvip",
            "TestDrives",
            "TestHardwareConfig",
            "TestKeyProviderKmip",
            "TestKeyServerKmip",
            "TestLdapAuthentication",
            "TestLocalConnectivity",
            "TestLocateCluster",
            "TestNetworkConfig",
            "TestPing",
            "TestRemoteConnectivity",
            "UpdateBulkVolumeStatus",
            "UpdateIdpConfiguration",
            "UpdateSnapMirrorRelationship"
        ],
        "currentVersion": "12.0",
        "supportedVersions": [
            "1.0",
            "2.0",
            "3.0",
```

```
            "4.0",
            "5.0",
            "5.1",
            "6.0",
            "7.0",
            "7.1",
            "7.2",
            "7.3",
            "7.4",
            "8.0",
            "8.1",
            "8.2",
            "8.3",
            "8.4",
            "8.5",
            "8.6",
            "8.7",
            "9.0",
            "9.1",
            "9.2",
            "9.3",
            "9.4",
            "9.5",
            "9.6",
            "10.0",
            "10.1",
            "10.2",
            "10.3",
            "10.4",
            "10.5",
            "10.6",
            "10.7",
            "11.0",
            "11.1",
            "11.3",
            "11.5",
            "11.7",
            "11.8",
            "12.0"
        ]
    }
}
```

GetAsyncResult

您可以使用 `GetAsyncResult` 检索异步方法调用的结果。某些方法调用需要一段时间才能运行，并且在系统发送初始响应时可能无法完成。要获取方法调用的状态或结果，请使用 `GetAsyncResult` 轮询方法返回的 asyncHandle 值。

`GetAsyncResult` 以标准方式返回操作的整体状态（正在进行，已完成或错误），但为操作返回的实际数据取决于原始方法调用，并且返回的数据会随每个方法记录在一起。

如果 keepResult 参数缺失或错误，则返回结果后 asyncHandle 将变为非活动状态，并且稍后尝试查询 asyncHandle 时会返回错误。通过将 keepResult 参数设置为 true ，您可以使 asyncHandle 保持活动状态以供将来查询。

Parameters

此方法具有以下输入参数：

Name	Description	Type	默认值	Required
异步处理	从原始异步方法调用返回的值。	整型	无	是的。
keepResult	如果为 true ，则 GetAsyncResult 不会在返回异步结果后将其删除，从而允许将来查询该 asyncHandle 。	boolean	false	否

返回值

此方法具有以下返回值：

Name	Description	Type
status	异步方法调用的状态。可能值： • running ： 此方法仍在运行。 • complete ： 此方法已完成，并且结果或错误可用。	string
结果	如果异步方法成功完成，则这是异步操作的结果。如果异步操作失败，则此成员不存在。	string

Name	Description	Type
error	如果状态为 complete 且异步方法失败，则此成员包含错误详细信息。如果异步操作成功，则此成员不存在。	string
resultType	异步方法调用正在执行或正在执行的操作类型。	string
详细信息	如果状态为 running ，则此成员包含与方法的当前操作相关的信息。如果异步方法未运行，则此成员不存在。	JSON 对象
CREATETIME	调用异步方法的时间，采用 UTC+0 格式。	ISO 8601 日期字符串
最后更新时间	上次更新异步方法状态的时间，采用 UTC+0 格式。	ISO 8601 日期字符串

- 注： * GetAsyncResult 的返回值实质上是标准 JSON 响应的嵌套版本，并具有一个附加的状态字段。

请求示例

此方法的请求类似于以下示例：

```
{
  "method": "GetAsyncResult",
  "params": {
      "asyncHandle" : 389
},
"id" : 1
}
```

响应示例：方法错误

此方法返回类似于以下示例的响应：

```
{
   "error": {
      "code": 500,
      "message": "DBClient operation requested on a non-existent path at
[/asyncresults/1]",
      "name": "xDBNoSuchPath"
   },
   "id": 1
}
```

如果 "response" 是 GetAsyncResult 调用中的 JSON 响应对象，则 "response.error" 将与 GetAsyncResult 方法本身的错误相对应（例如，查询不存在的 asyncHandle ）。

响应示例：异步任务错误

此方法返回类似于以下示例的响应：

```
{
   "id": 1,
   "result": {
     "createTime": "2016-01-01T02:05:53Z",
     "error": {
       "bvID": 1,
       "message": "Bulk volume job failed",
       "name": "xBulkVolumeScriptFailure",
       "volumeID": 34
     },
     "lastUpdateTime": "2016-01-21T02:06:56Z",
     "resultType": "BulkVolume",
     "status": "complete"
   }
}
```

`reonse.result.error` 对应于原始方法调用的错误结果。

响应示例：异步任务成功

此方法返回类似于以下示例的响应：

```
{
   "id": 1,
   "result": {
     "createTime": "2016-01-01T22:29:18Z",
     "lastUpdateTime": "2016-01-01T22:45:51Z",
     "result": {
       "cloneID": 25,
       "message": "Clone complete.",
       "volumeID": 47
     },
     "resultType": "Clone",
     "status": "complete"
   }
}
```

`reonse.result.result` 是在调用成功完成时原始方法调用的返回值。

自版本以来的新增功能

9.6

GetCompleteStats

NetApp 工程人员使用 `GetCompleteStats` API 方法测试新功能。从 `GetCompleteStats` 返回的数据不会进行记录，并且会频繁更改，并且无法保证其准确性。您不应使用 `GetCompleteStats` 收集性能数据或与运行 Element 软件的存储集群进行任何其他管理集成。

使用以下受支持的 API 方法检索统计信息：

- GetVolumeStats
- GetClusterStats
- GetNodeStats
- GetDriveStats

自版本以来的新增功能

9.6

GetLimits

您可以使用 `GetLimits` 方法获取 API 设置的限制值。这些值可能会在 Element 版本之间发生更改，但如果不更新系统，这些值不会发生更改。在为面向用户的工具编写 API 脚本时，了解 API 设置的限制值会很有用。

无论用于传递此方法的 API 端点版本如何， `GetLimits` 方法均返回当前软件版本的限制。

Parameters

此方法没有输入参数。

返回值

此方法将返回一个 JSON 对象，其中包含名称 - 值对，这些名称 - 值对包含 API 限制。

请求示例

此方法的请求类似于以下示例：

```
{
   "method": "GetLimits",
   "id" : 1
}
```

响应示例

此方法返回类似于以下示例的响应：

```
{
    "id": 1,
    "result": {
        "accountCountMax": 5000,
        "accountNameLengthMax": 64,
        "accountNameLengthMin": 1,
        "backupTargetNameLengthMax": 64,
        "backupTargetNameLengthMin": 1,
        "bulkVolumeJobsPerNodeMax": 8,
        "bulkVolumeJobsPerVolumeMax": 2,
        "chapCredentialsCountMax": 15000,
        "cloneJobsPerNodeMax": 8,
        "cloneJobsPerVirtualVolumeMax": 8,
        "cloneJobsPerVolumeMax": 2,
        "clusterAdminAccountMax": 5000,
        "clusterAdminInfoNameLengthMax": 1024,
        "clusterAdminInfoNameLengthMin": 1,
        "clusterPairsCountMax": 4,
        "fibreChannelVolumeAccessMax": 16384,
        "initiatorAliasLengthMax": 224,
        "initiatorCountMax": 10000,
        "initiatorNameLengthMax": 224,
```

```
        "initiatorsPerVolumeAccessGroupCountMax": 128,
        "iscsiSessionsFromFibreChannelNodesMax": 4096,
        "maxAuthSessionsForCluster": 1024,
        "maxAuthSessionsPerUser": 1024,
        "nodesPerClusterCountMax": 100,
        "nodesPerClusterCountMin": 3,
        "qosPolicyCountMax": 500,
        "qosPolicyNameLengthMax": 64,
        "qosPolicyNameLengthMin": 1,
        "scheduleNameLengthMax": 244,
        "secretLengthMax": 16,
        "secretLengthMin": 12,
        "snapMirrorEndpointIPAddressesCountMax": 64,
        "snapMirrorEndpointsCountMax": 4,
        "snapMirrorLabelLengthMax": 31,
        "snapMirrorObjectAttributeValueInfoCountMax": 9900000,
        "snapshotNameLengthMax": 255,
        "snapshotsPerVolumeMax": 32,
        "storageNodesPerClusterCountMin": 2,
        "virtualVolumeCountMax": 8000,
        "virtualVolumesPerAccountCountMax": 10000,
        "volumeAccessGroupCountMax": 1000,
        "volumeAccessGroupLunMax": 16383,
        "volumeAccessGroupNameLengthMax": 64,
        "volumeAccessGroupNameLengthMin": 1,
        "volumeAccessGroupsPerInitiatorCountMax": 1,
        "volumeAccessGroupsPerVolumeCountMax": 64,
        "volumeBurstIOPSMax": 200000,
        "volumeBurstIOPSMin": 100,
        "volumeCountMax": 4000,
        "volumeMaxIOPSMax": 200000,
        "volumeMaxIOPSMin": 100,
        "volumeMinIOPSMax": 15000,
        "volumeMinIOPSMin": 50,
        "volumeNameLengthMax": 64,
        "volumeNameLengthMin": 1,
        "volumeSizeMax": 17592186044416,
        "volumeSizeMin": 1000000000,
        "volumesPerAccountCountMax": 2000,
        "volumesPerGroupSnapshotMax": 32,
        "volumesPerVolumeAccessGroupCountMax": 2000,
        "witnessNodesPerClusterCountMax": 4
    }
}
```

自版本以来的新增功能

9.6

GetOrigin

您可以使用 `GetOrigin` 方法获取构建节点的源证书。

Parameters

> 如果没有源证书，则此方法将返回 "null" 。

此方法没有输入参数。

返回值

此方法将返回供应商原始认证信息。

请求示例

此方法的请求类似于以下示例：

```
{
 "method": "GetOrigin",
 "id" : 1
}
```

响应示例

此方法返回类似于以下示例的响应：

```
{
  "integrator": "SolidFire",
  "<signature>": {
    "pubkey": [public key info],
    "version": 1,
    "data": [signature info]
  },
   "contract-id": "none",
   "location": "Boulder, CO",
   "organization": "Engineering",
   "type": "element-x"
  }
 ]
}
```

自版本以来的新增功能

9.6

GetRawStats

NetApp 工程人员使用 `GetRawStats` API 方法测试新功能。从 `GetRawStats` 返回的数据不会进行记录，并且会频繁更改，并且无法保证其准确性。您不应使用 `GetRawStats` 来收集性能数据或与运行 Element 软件的存储集群进行任何其他管理集成。

使用以下受支持的 API 方法检索统计信息：

- GetVolumeStats
- GetClusterStats
- GetNodeStats
- GetDriveStats

自版本以来的新增功能

9.6

ListAsyncResults

您可以使用 `ListAsyncResults` 列出系统上所有当前正在运行和已完成的异步方法的结果。使用 `ListAsyncResults` 查询异步结果不会使发生原因已完成的 asyncHandles 过期；您可以使用 `GetAsyncResult` 查询 `ListAsyncResults` 返回的任何 asyncHandles 。

参数

此方法具有以下输入参数：

Name	Description	Type	默认值	Required
asyncResultTypes	结果类型的可选列表。您可以使用此列表将结果限制为仅包含这些类型的操作。可能值： • DriveAdd ：涉及系统向集群添加驱动器的操作。 • BulkVolume ：卷之间的复制操作，例如备份或还原。 • clone ：卷克隆操作。 • DriveRemoval ：涉及系统从驱动器复制数据以准备将其从集群中删除的操作。 • RtfiPendingNode ：涉及系统在将兼容软件添加到集群之前在节点上安装该软件的操作。	string array	无	否

返回值

此方法具有以下返回值：

Name	Description	Type
异步处理	序列化异步方法结果数组。	JSON 对象数组

请求示例

此方法的请求类似于以下示例：

```
{
   "method": "ListAsyncResults",
   "params": {
   },
   "id": 1
}
```

响应示例

此方法返回类似于以下示例的响应：

```
{
   "id": 1,
   "result": {
     "asyncHandles": [
       {
         "asyncResultID": 47,
         "completed": true,
         "createTime": "2016-01-01T22:29:19Z",
         "data": {
           "cloneID": 26,
           "message": "Clone complete.",
           "volumeID": 48
         },
         "lastUpdateTime": "2016-01-01T22:45:43Z",
         "resultType": "Clone",
         "success": true
      },
      ...]
   }
}
```

自版本以来的新增功能

9.6

了解更多信息

GetAsyncResult

版权信息

版权所有©2022 NetApp、Inc.。保留所有权利。Printed in the U.S.版权所涵盖的本文档的任何部分不得以任何形式或任何手段复制、包括影印、录制、 磁带或存储在电子检索系统中—未经版权所有者事先书面许可。

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

本软件由NetApp按"原样"提供、不含任何明示或默示担保、包括但不限于适销性和特定用途适用性的默示担保、特此声明不承担任何责任。IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice.NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp.The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S.patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

商标信息

NetApp、NetApp标识和中列出的标记 http://www.netapp.com/TM 是NetApp、Inc.的商标。其他公司和产品名称可能是其各自所有者的商标。